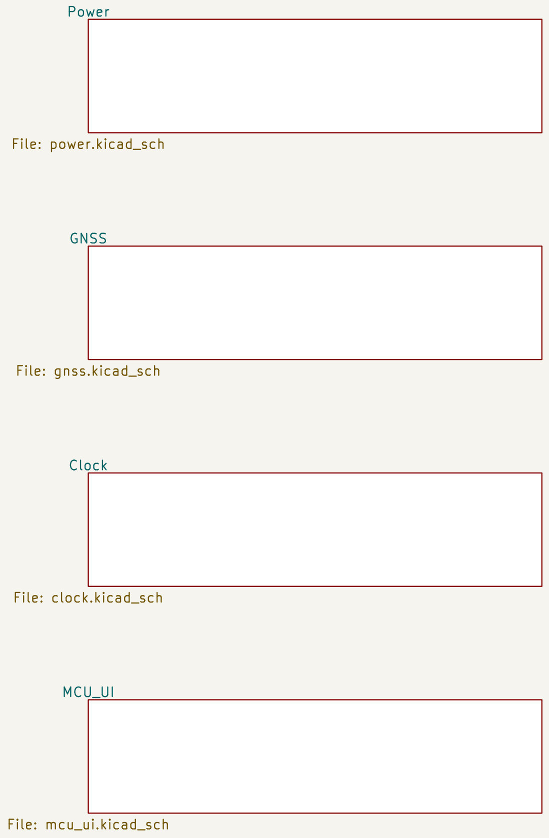


Sheet: /
File: hamref_proto_v0_1.kicad_sch

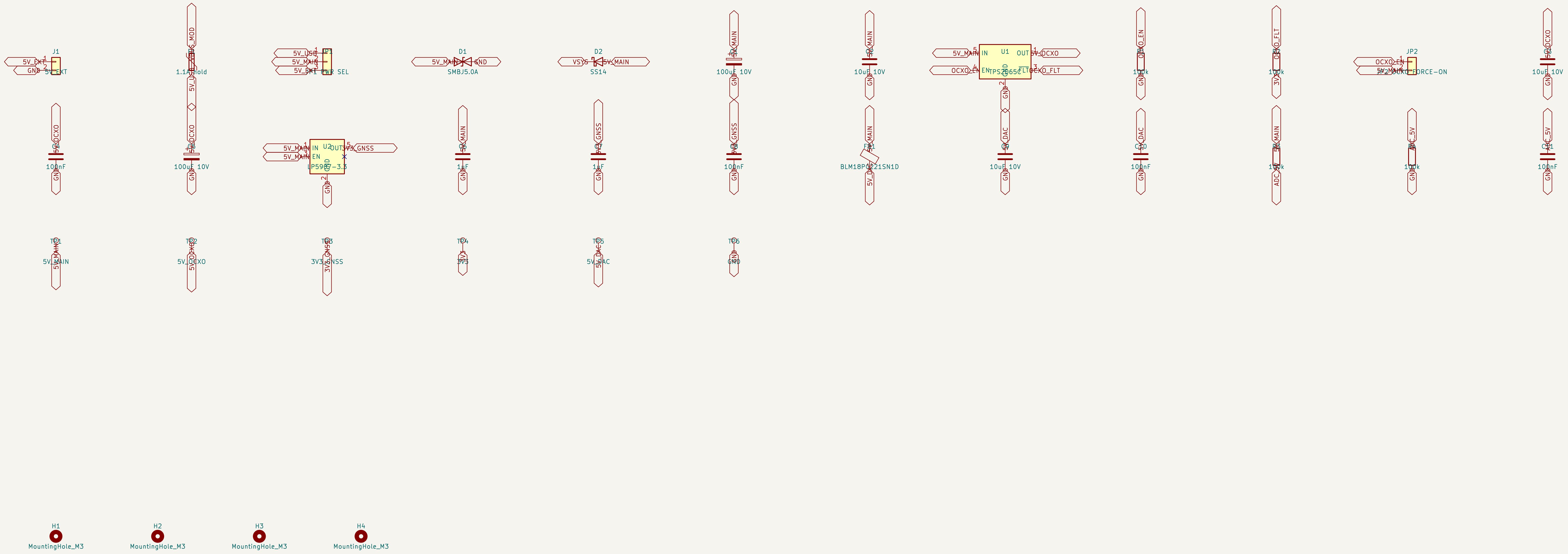
Title:

Size: A3 Date: **Rev:**

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 1/5

POWER: 5V_USB (VBUS) / 5V_EXT -> JP1 -> 5V_MAIN -> TVS, OCXO switch, 3V3_GNSS LDO, ferrite-filtered 5V_DAC

JP1: 1-2 = USB power, 2-3 = external 5V. Module VSYS fed from 5V_MAIN through Schottky (D2). VBUS pad is direct USB VBUS.



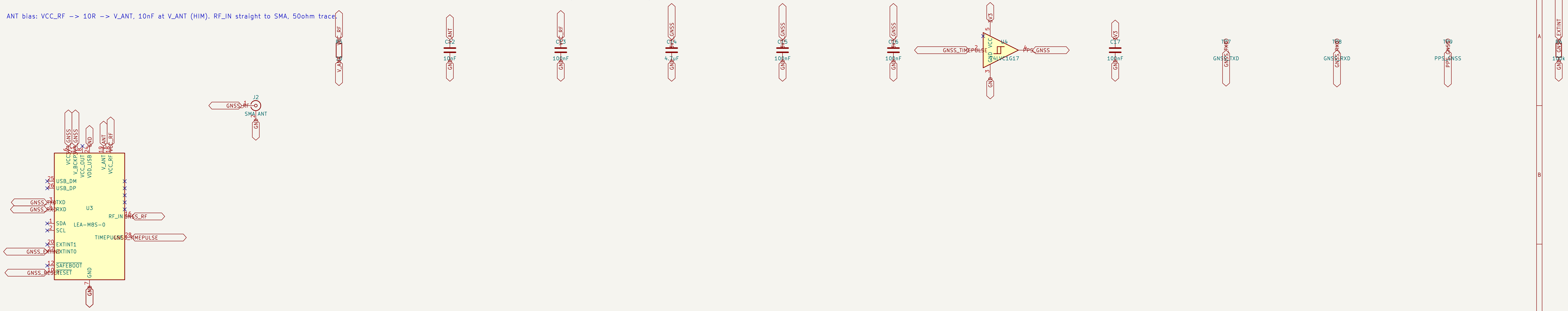
Sheet: /Power/
File: power.kicad_sch

Title:

Size: A2
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 2/5

GNSS: u-blox LEA-M8S on 3V3_GNSS. UART to MCU, TIMEPULSE -> 74LVC1G17 -> GP6. USB not used (VDD_USB=GND).

ANT bias: VCC_RF -> 10R -> V_ANT, 10nF at V_ANT (HIM). RF_IN straight to SMA, 50ohm trace.



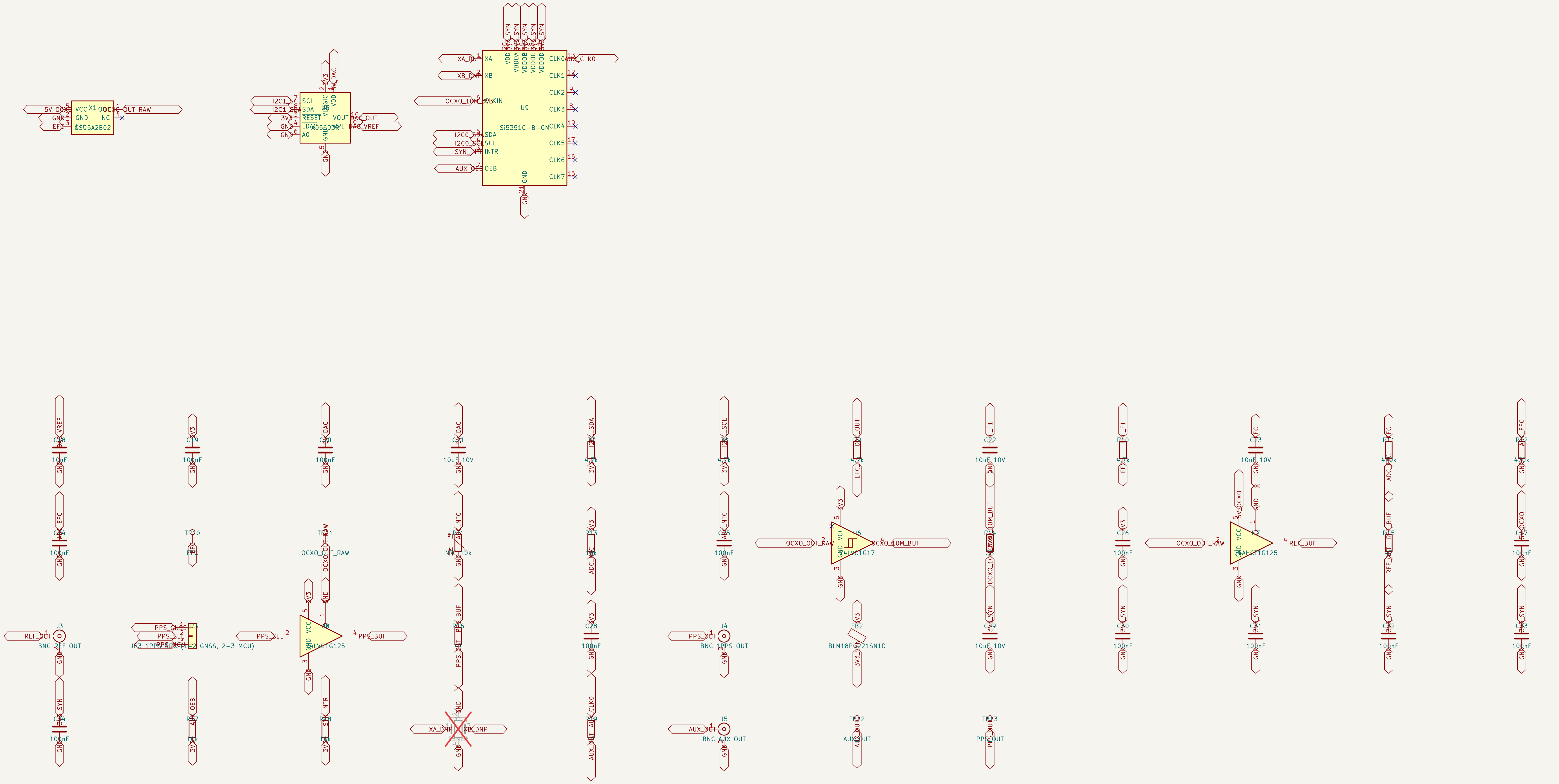
Sheet: /GNSS/
File: gnss.kicad_sch

Title:

Size: A2
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 3/5

CLOCK: OCXO + EFC DAC, 10MHz buffers, 1PPS OUT mux, Si5351C AUX synth (CLKIN = OCXO). I2C1: DAC 0x4C. I2C0: Si5351 0x60 + OLED 0x3C.

OCXO pin geometry unverified: placeholder 1x5 header, pad N = OCXO pin N (datasheet bottom view). Verify before layout.

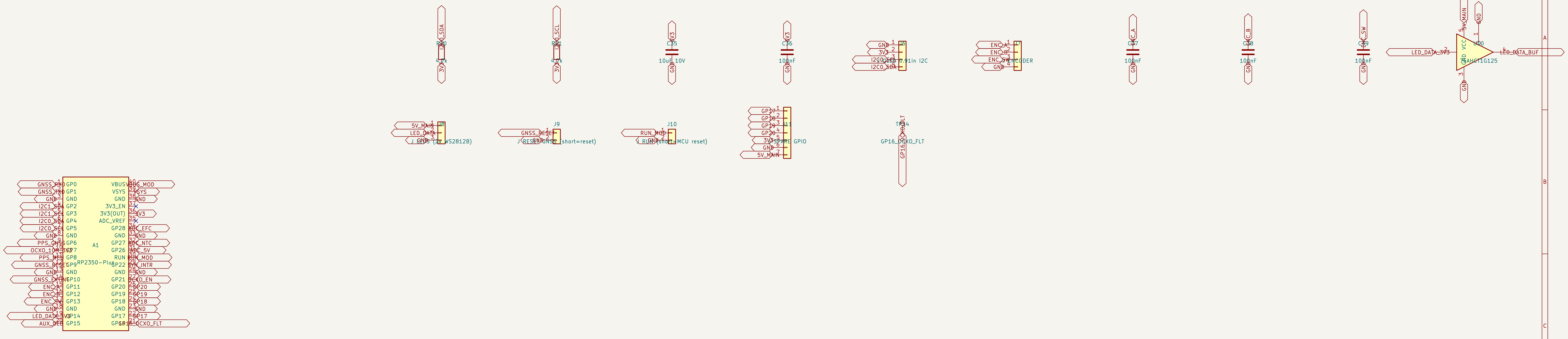


Sheet: /Clock/
File: clock.kicad_sch

Title:

Size: A2	Date:	**Rev:**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/5

MCU + UI: Waveshare RP2350-Plus module soldered via castellated pads. GPIO map = docs v0.9 (GP16 = OCXO_FLT).



Sheet: /MCU_UI/
File: mcu_ui.kicad_sch

Title:

Size: A2
Date:
Rev:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 5/5